

Aditya Sunke

Blacksburg, VA | [Email](#) | [LinkedIn](#) | [GitHub Profile](#)

EDUCATION

Virginia Tech, Blacksburg, VA

Expected Graduation: May 2027

Major: B.S. Computer Science (C.S.) | **Minor:** Quantum Information Science and Engineering (Q.I.S.E.)

GPA: 3.81/4.0 | **Dean's List:** Fall 2023, Spring, 2024, Fall 2024, Spring 2025

Relevant Coursework: Quantum Software, Quantum World, Quantum Information Technologies, Quantum Computation, Linear Algebra, Data Structures and Algorithms, Software Design, Computer Organization, Combinatorics, Graph Theory

WORK EXPERIENCE

Co-founder and Quantum Lead

July 2025 – Present

CarbonQapture

Blacksburg, VA

- Co-founded and led a 7-member interdisciplinary team focused on applying **quantum computing and AI** to carbon capture technologies, securing over **\$300 in university funding** each semester to support **R&D** efforts.
- Designing and implementing a **hybrid VQE-SQD algorithm** for **quantum chemistry simulations** in **Qiskit and Cirq** to simulate Carbon Dioxide with Metal Organic Frameworks enabling **near-term quantum application** testing.

Research Assistant

Mar 2025 – July 2025

National Science Foundation

Blacksburg, VA

- Advanced cryptographic research under a **\$247K NSF-funded grant** by analyzing multivariate Goppa code properties and prototyping their integration into the McEliece cryptosystem, strengthening **resilience against quantum attacks**.
- Integrated **multivariate Goppa codes** into the McEliece cryptosystem **improving decoding efficiency** while validating resilience against quantum attacks, contributing to **NIST** standardization efforts.

AI Developer

Jan 2025 – May 2025

Virginia Tech College of Engineering

Blacksburg, VA

- Collaborated within an **interdisciplinary team** to design an **AI-assisted navigation system** enabling **visually impaired individuals** to explore tourist and museum spaces independently.
- Developed and deployed a **prototype wearable device** (Raspberry Pi-based necklace) integrating **Microsoft Azure Computer Vision** and a custom-trained **Google Gemini model** on museum artifacts for real-time environmental understanding.

Undergraduate Student Researcher

Dec 2024 – Mar 2025

Virginia Tech

Blacksburg, VA

- Researched the McEliece cryptosystem for **post-quantum encryption**, focusing on security resilience against quantum algorithms and implemented the McEliece cryptosystem in SageMath to validate performance and theoretical resistance.
- Verified Patterson's algorithm for decoding Goppa codes up to **4096-bit** key lengths, demonstrating real-world applications.

PROJECTS

[SilentVoice](#)

Dec 2025 – Present

- Developing a **computer vision and machine learning system** to **translate sign language** into natural language sentences.
- Developing a **CNN-based finger-spelling recognition model** to classify hand gesture images into alphabet characters, enabling **real-time translation of sign language** finger spelling into text.

[Qure AI](#)

Sept 2025

- Developed an **SVM-based drug classification model** achieving **90%+ prediction accuracy** across four meningitis-causing bacterial datasets and applied **Variational Quantum Eigensolver (VQE)** algorithm to simulate ground-state energies.
- Generated **22 novel drug candidates** with **ground-state energies 1.98× lower** than existing drugs, indicating higher stability.

[Q-Folio](#)

July 2025

- Implemented **QAOA** using **IBM Qiskit** to optimize stock portfolios, demonstrating practical performance of near-term quantum algorithms on real-world datasets to optimize stock portfolios using **100 days of market activity** from **Alpha Vantage API**.
- Achieved **~20% reduction in portfolio variance** compared to brute-force, demonstrating **reduced risk in stock selection**.

TECHNICAL SKILLS

Languages | Python, Java, C/C++, SQL, x86_64 Assembly

Libraries | Pandas, NumPy, Matplotlib, SageMath, SciPy, PySCF, PyTorch

Developer Tools | Git, GitHub, Docker, Visual Studio Code, PyCharm, Eclipse, Jupyter, Linux, Hugging Face

Quantum Development | Qiskit, Cirq, PennyLane, AWS Braket, IBM Quantum Cloud, Quantum Chemistry, Post-Quantum Cryptography, Quantum Finance, Quantum and Hybrid Algorithms (VQE, QAOA)